Arabic Sentiment Analysis Using Recurrent Neural Networks: A Review

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Abstract
Over the last decade, the amount of Arabic content created on websites and social media has grown significantly. Opinions are shared openly and freely on social media and thus provide a rich source for trend analyses, which are accomplished by conventional methods of language interpretation, such as sentiment analysis. Due to its accuracy in studying unstructured data, deep learning has been increasingly used to test opinions. Recurrent neural networks (RNNs) are a promising approach in textual analysis and exhibit large morphological variations. In total, 193 studies used RNNs in English-language sentiment analysis, and 24 studies used RNNs in Arabic-language sentiment analysis. Those studies varied in the areas they address, the functionality and weaknesses of the models, and the number and scale of the available datasets for different dialects. Such variations are worthy of attention and monitoring; thus, this paper presents a systematic examination of the literature to label, evaluate, and identify state-of-the-art studies using RNNs for Arabic sentiment analysis.

Keywords Sentiment analysis · Deep learning · Opinion mining · Sentiment classification · Arabic language · Recurrent neural network

1 Introduction

The Internet's strength and reach continue to expand globally. According to the International Data Corporation, the amount of digital data generated worldwide exceeded 33 zettabytes in 2018, with a projected growth of 175 zettabytes by 2025 (Reinsel et al. 2018). The number of Internet users worldwide has risen to 4,021 billion users; in particular, the number of Middle East users has increased to 164 million users (Kemp 2018). In recent years, an increasing number of people have used social media to share their views or leave a positive or negative review of a specific service. As of January 2018, 3,196 billion people are on social media, and 130 million of them are Arabs (Kemp 2018).
Due to the influence of social media content on governing, development, diplomacy, and business, sentiment analysis is required in social media monitoring, as it allows an overview of the wider public opinion on topics appearing in a variety of posts, from politics-related posts to customer reviews. The ability to discern the sentiment and attitude behind a post on any subject facilitates strategizing and planning, thus providing better services.

Sentiment analysis is the computational assessment of a nation's attitudes, feelings, and emotions towards structures, persons, cases, news, or subjects. Understanding open

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opinions, thoughts, and questions expressed is a matter of urgency at this point, which is the reason for excitement regarding sentiment analysis. A considerable amount of research has been conducted to improve the precision of sentiment analysis, from basic linear approaches to more complex deep neural network models (Heikal et al. 2018).

Machine learning techniques have been commonly used in sentiment analysis. However, these techniques have limited ability to process raw data, and feature representation greatly affects the performance of a machine learning model. For this reason, deep learning is used for feature representation at multiple levels. Deep learning automatically discovers discriminative and explanatory text representations from data using nonlinear neural networks, each of which transforms the representation at one level into a representation at a higher and more abstract level (LeCun et al. 2015).

Recently, deep learning has been shown to be highly productive in sentiment analysis and is considered a modern multilingual model for sentiment analysis. The analysis of Arabic sentiments, however, still needs improvement. Due to its complex structure and different dialects, as well as a lack of resources, Arabic language processing faces many challenges. While current deep learning approaches have enhanced the accuracy of sentiment analysis in Arabic, these approaches can still be improved (Heikal et al. 2018).

A promising new research field in Arabic sentiment analysis is the application of recurrent neural networks (RNNs) in textual models to demonstrate the learning process and measure understanding of the text at the level of semantic analysis (Souri et al. 2018).

In this study, we aim to highlight and discuss the research body on Arabic sentiment analysis using RNNs for the following reasons:

First, RNNs vs. machine learning models: As machine learning models rely on feature representation, the goodness of the data representation is directly proportional to the performance gains (Najafabadi et al. 2015). Since there is no correct way to represent these features, it is domain-specific, costly and requires much preprocessing, which is especially true for the Arabic language with its multiple dialects. Moreover, upon analyzing the literature on Arabic sentiment analysis, we found that RNNs for Arabic sentiment analysis always outperformed machine learning models for Arabic sentiment analysis except in only one study where an SVM outperformed an RNN because of the use of rich handcrafted feature sets (morphological, N-grams, syntactic, and semantic features) (Al-Smadi et al. 2018). Furthermore, RNNs do not require domain expertise or hardcore feature extraction, as they learn high-level features from data in an incremental manner. The main reason is due to the rich sequence-to-sequence model of RNNs.

Second, RNNs vs. other deep learning models: A related study (Yadav and Vishwakarma 2019) indicated that RNNs are one of the best and most important methods for text-based sentiment analysis, while convolutional neural networks (CNNs) have shown good results for image processing tasks such as image sentiment analysis. Based on a comparison between an RNN and a CNN for natural language processing (Yin et al. 2017), the CNN outperformed the RNN in specific tasks (answer selection and question relation matching), while the RNN showed promising results in the remaining natural language processing tasks (sentiment classification, relation classification, textual entailment, path query answering and part-of-speech tagging), especially in the sentiment classification tasks because RNNs are trained to recognize temporal patterns, while CNNs learn to recognize spatial patterns. In textual analysis, CNNs can identify special patterns of n-gram words regardless of the positions and orders of these words (Yin et al. 2017), which leads to ignoring the semantic dependency and losing information. Hence, the integration depth of a CNN is not sufficient to process the language complexity. On the other hand, RNNs are inherently temporally deep, since their hidden state is a function of all previous hidden states, which means it is more suitable to capture sequential data. To enhance an RNN to capture spatial depth, we could add multiple hidden layers, which means that we could use an RNN alone to achieve spatiotemporal depth integration (Hermans and Schrauwen 2013).

In general, RNNs are crucial in sentiment analysis, whether used alone or with other architectures, because they are specifically designed for text classification tasks. Despite the importance and effectiveness of RNNs in sentiment analysis for various languages, this aspect has not been studied for Arabic. Therefore, in this study, RNNs are the focal point of the review to analyze and compare different studies on effective Arabic sentiment analysis.

In Section 1.1, after presenting an overview of sentiment analysis, its importance and the reason for choosing RNNs as the focal point, we discuss Arabic sentiment analysis across different studies and explain why sentiment analysis for Arabic is slightly difficult until now. In addition, we discuss the complexity of the Arabic language in Section 1.2. This section discusses the diversity of Arabic forms, orthography, and morphology and how these challenges affect language processing. Moreover, we present the contributions of this paper in Section 1.3.

1.1 Sentiment analysis for Arabic

Sentiment analysis or opinion mining is a computational method for the classification of thoughts, emotions, and attitudes regarding goods, services, or subjects expressed using text. This analysis is achieved by defining and categorizing these opinions and using natural language processing techniques to derive subjective knowledge (Medhat et al. 2014).

A large amount of unstructured data has been generated daily since the 2000s with the rise of the World Wide Web and the Internet. This amount has started to grow more rapidly with the advent of social media and smartphones, which create big data. Since then, sentiment analysis has become an important natural language processing tool to grasp trends and classify opinions. It is widely used to mine reviews, survey responses, and social media to evaluate services and improve productivity in different areas, such as politics, commerce, tourism, education, and health (Zhang et al. 2018). sentiment analysis involves collecting reliable data from various resources, identifying the opinion embedded in each phrase, preparing labeled data for analysis, and extracting features by classifying these phrases (Hemmatian and Sohrabi 2017; Rani and Kumar 2019).

In (Alsayat and Elmitwally 2020), the different components present in the Arabic language are explained. The Arabic sentiment analysis involves the following components: phonetics, syntax, lexicology, morphology, semantics, etc. The most important factor in English and the Indo-European languages is the opinions and the feelings present in the phrases. However, in the Arabic language, little emphasis is given to these factors. In the proposed Arabic sentiment analysis framework, rhetorical characteristics and implicit meanings are given more priority over the linguistic levels mentioned above. Due to the limitations of Arabic sentiment analysis research, the scope of developments to build a better sentiment analysis engine is in progress (Shoukry and Rafea 2012). The importance of the application of the levels that determine the sentiment in tweets is also a challenge that is explained in the paper (Abdulla et al. 2013). This part of the sentiment, once labeled across the different tweets, is then used as a reference dataset that can then be analyzed to give the sentiment of incoming tweets in real time. In this paper, the model giving the highest recall was considered the best algorithm. The term recall will be discussed in detail later in our research paper. The developed models were trained and validated on the tweets and the data generated in-house to see the validity of the model generation scheme and performance.

1.2 Arabic sentiment analysis challenges

We can categorize Arabic sentiment analysis challenges into three groups: language forms, language orthography and language morphology (Saleh 2009; Habash 2010; Turki Khemakhem et al. 2010; Boudad et al. 2018). These groups are explained in greater detail as follows:

(i) **Arabic forms:** Arabic has three primary forms: Classical Arabic, modern standard Arabic (MSA), and dialectical Arabic. The Qur'an, the holy book of Islam, is in classical Arabic. MSA is similar to classical Arabic with less sophistication and more generic words; it is used in formal written and spoken media such as the news, education, and literature. Dialectical or colloquial Arabic is used mostly in daily life and has regional variations. Although dialectical Arabic is used mainly as a spoken language, currently, it is used in written social communication on social media and short messages and includes more than 30 dialects (Habash 2010). Dialects are generally classified into six basic groups (Habash 2010): Egyptian (including Egypt and Sudan), Levantine (including Jordan, Syria, Palestine, and Lebanon), Gulf (including Saudi Arabia, Oman, Kuwait, United Arab Emirates, Qatar, and Bahrain), North African (including Morocco, Tunisia, Algeria, Libya, and Mauritania), Iraqi, and Yemeni. For example, the expression **كيف حالك** (how are you) in MSA has different forms in each dialect (Egyptian: **إزيك**, Levantine: **كيفك**, Gulf: **شلونك**, North African: **شنوه احوالك**, Iraqi: **اشونك**, Yemeni: **كيفوك**).

(ii) **Arabic orthography:** Arabic text is written from right to left and is characterized by the absence of upper or lower cases. Its alphabet contains 28 letters: 25 consonants and only 3 vowels (**أ, و,ي**). In addition, short vowels (ُ َ ِ) are used as diacritical marks and are put either over or under letters to indicate the exact articulation and to explain the meaning of the text. Moreover, based on the presence or absence of such diacritics, the meaning of words can be different. For example, the word (**علم**) may mean **عِلْم** (knowledge), **عَلَم** (flag), or **عَلَّمَ** (teach), wherein the letters are identical while the diacritics are different. In the form of written Arabic used on social media, most words are written without diacritics, which makes it even more challenging to analyze (Boudad et al. 2018).

(iii) **Arabic morphology:** The Arabic language has a highly complicated morphology, in which a word may carry significant information. A term in Arabic has several morphological characteristics as a spatially delimited token: agglutinative, inflectional, and derivational morphology (Saleh 2009; Turki Khemakhem et al. 2010; Boudad et al. 2018). The morphological features can be described as follows:

- Agglutinative morphology: Compared to nonagglutinative languages, the number of possible words produced by different morphemes of a single root is high. A word in Arabic can consist of a stem plus one or more affixes and clitics. A stem is a combination of a root and derived morphemes that can be attached to one or more affixes. On the other hand, the clitics include proclitics, which occur at the beginning of a word, such as the letters (**و, ف**) meaning (and, then), and enclitics that occur at the ending of a word, which are complement pronouns. For example, the Arabic expression (**ليكتبونـها**) meaning (for them to write it) contains several attached affixes and clitics as follows: (**ل** proclitic), (**ي** prefix), (**كتب** stem), (**ون** suffix), and (**ها** enclitic). The complex nature of Arabic morphology leads to analytical ambiguity. For example, the word (**وجدنا** we found) can be analyzed as (**وجد** found + **نا** we) and as (**و** and + **جد** grandfather + **نا** we).
- Derivational morphology: Derivation is a method of deriving a new word from an existing word, involving the alteration of a part-of-speech, a change in meaning, or both. For example, from the root (**ك ت ب**) that constructs the verb (**كَتَبَ**) meaning (wrote), one can derive five or more derivations such as (**كَاتِب** writer), (**مَكْتُوب** letter), (**كِتَابَةٌ** writing), (**كِتَاب** book), and (**مَكْتَب** office); thus, one root can generate different words with different meanings, which increases ambiguity when analyzing the language.

- Inflectional morphology: Inflection is the variation or change in grammatical form that words undergo to mark distinctions of case, gender, number, and person. For example, the variation in gender can be noticed in the word (student), which can be written as (**طالب**) for males and (**طالبة**) for females. Another word (white) can be written as (**أبيض**) for males and (**بيضاء**) for females. This group of inflected word shapes is called a lexeme category. To characterize a lexeme, a lemma, which is an exact shape, is conservatively chosen. The diversity of inflected Arabic words leads to a major challenge in natural language processing.

The diversity of dialects of the Arabic language, along with the richness in scripts, orthography, morphology, phonology, and semantics, poses research challenges that require appropriate systems and solutions to handle the ambiguity through the application of tokenization, spelling checks, stemming, lemmatization, pattern matching, and part-of-speech tagging. Extraction techniques are the backbone affecting model performance. In other words, the better the feature extraction technique is, the better the sentiment analysis. Feature extraction in different languages impacts the sentiment analysis results and can be carried out in multiple ways. Feature extraction is mainly concerned with the task of finding and yielding different statistical and transfer functions for important words in a dataset, which is a well-defined procedure in English compared to other languages (Dashtipour et al. 2016). In addition, this study mentioned that the main difference of this approach is the seed corpus, which is prepared by delimiting a sequence of characters by noncharacter symbols, by counting the number of occurrences that follow a negated adverbial phrase, by counting the number of unconstructed sequences, etc. In a review of sentiment analysis carried out in the domain of the Arabic language, an in-depth analysis was performed on the approaches related to sentiment analysis (Oueslati et al. 2020). To create a corpus for machine learning approaches, a hybrid-based method is proposed, which tries to combine corpus-based and lexicon-based approaches, that focuses on the domain adaptation of sentiments and addresses poor language resources by using annotated corpora and lexicon resources. The polarity classification step creates a structured dataset that can be used for model building purposes and is one of the bases of any sentiment analysis paper (Oueslati et al. 2020).

1.3 Contributions

The contributions of this survey are summarized in Figure 1 and as follows.

- A systematic review technique is implemented to identify recent RNN models for sentiment analysis.
- The annotated datasets for Arabic languages, which are available online, are discussed for further research.
- The research studies are explained, categorized by sentiment type (emoji analysis, emotion detection, sentiment classification, or hate speech detection) based on the level (sentence, document, or aspect).
- The limitations and drawbacks of the methods are indicated so that alternative approaches may be used.
- Future research directions are discussed in the last section.

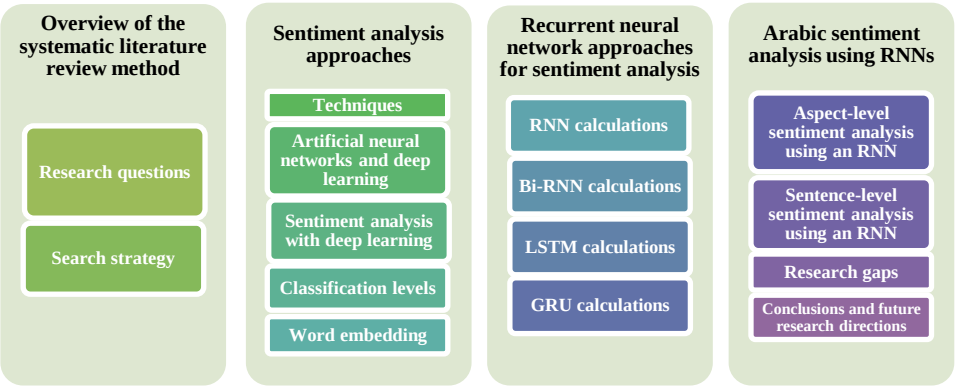

Fig. 1. Survey framework

In Section 2, we present an overview of the systematic literature review method, including the four research questions and the strategies used to extract the targeted studies for both English and Arabic. In Sections 3, 4, and 5, we answer the first, second, and third research questions, respectively. In Section 6, we describe the evaluation methods for the studies under consideration. Finally, in Section 7, we answer the fourth question and conclude our study.

2 Overview of the systematic literature review method

This review is divided into two parts: RNNs for English text and RNNs for Arabic text. The systematic review follows the guidelines described by Kitchenham (Kitchenham 2004) and Heckman (Heckman and Williams 2011). The methodology is divided into the following sections: the research questions and the search strategy for both parts of the review. The research questions are outlined in Section 2.1. The search strategy is described in Section 2.2. The sole purpose of this research work is to check the feasibility of sentiment analysis on Arabic text data in different forms.

2.1 Research questions

Through this study, the following research questions (RQs) are of interest to us. RQ1 (sentiment analysis approaches), RQ2 (RNN approaches for sentiment analysis), RQ3 (related works on Arabic sentiment analysis using RNNs) and RQ4 (research gaps) will be discussed in Sections 3, 4, 5, and 7, respectively. Table 1 presents each RQ with its motivation.

Table 1. Research questions of the systematic review with their motivations

Definition	Motivation
RQ1: What are neural networks and how are these in sentiment analysis?	To provide background information for analytical techniques used in sentiment analysis.
RQ2: What are the types of RNNs for sentiment analysis?	To explore the state-of-the-art techniques and implementations of RNNs for sentiment analysis. We focus on recurrent LSTM and GRUs.
RQ3: What studies on Arabic sentiment analysis use RNNs?	To review all latest the models, techniques, and lexical resources for recurrent models used in Arabic sentiment analysis.
RQ4: What are the limitations and drawbacks found in these studies?	To investigate the gaps of recurrent models used in Arabic sentiment analysis.

2.2 Search strategy

In the first study, we used RNNs. We then removed those who did not use English for sentiment classification and compared the remaining studies with those who used Arabic.

We focus on English, as English is the dominant language for studies, given the abundance of datasets and available supportive tools. In the second part, we consider studies on Arabic sentiment analysis, as this is the focus of our research, and study potential gaps and challenges. The two parts have different keywords and elimination strategies, as demonstrated in the following sections.

(i) **RNNs for sentiment analysis:** We used the search sentence "x y" after many searches and observations to find the correct keywords, where x is "recurrent" and "neural," while y is "feeling" or "opinion." That is, we concentrate on all studies that use RNNs to evaluate sentiment by combining x and y, resulting in two possible search sentences. We searched the following databases: Springer, IEEE, ACM, Science Direct and other databases, namely, Aclweb, NIPS, AAAI, and Semantic Scholar. The search period was from 2013 through 2018. The year 2013 was chosen because no studies using RNNs for sentiment analysis prior to 2013 were found. The initial search resulted in 38,926 papers. The search aimed to cover all related studies with names, abstracts, keywords and complete texts. These papers were checked according to the following for final selection criteria: papers related to artificial intelligence were included, whereas irrelevant papers (such as those involving non-English sentiment analysis) were excluded. In addition, those concerned with nontarget areas, such as image analysis, video analysis, and gender identification, were excluded as well. After applying the selection criteria, there were 193 papers. Table 2 shows the total number of papers for each database using the mentioned keywords for RNNs in English.

Table 2. Number of studies found by the initial search for each keyword for all databases

Databases Keyword	Springer	IEEE	Science Direct	ACM	Other
Recurrent + neural + sentiment	691	859	325	524	11,600
Recurrent + neural + opinion	2,238	953	1,870	466	19,400
Total (initial search)	2,929	1,812	2,195	990	31,000
Total (after selection)	37	52	12	32	60

(ii) **RNNs for Arabic sentiment analysis:** In this part of the search, the focus was on studies concerned with RNNs for Arabic sentiment analysis, and we used the combination of three search terms. The first two terms were the same x and y as in the previous section, and the additional term “Arabic” was added to the two search sentences. We searched the same databases as in the previous section. Initially, the keyword search resulted in 2,636 papers from 2013 through 2018. The search targeted titles, abstracts, keywords, and full texts to cover all relevant studies. The papers were reviewed for final selection according to the criteria used in the previous section. In addition, papers on non- Arabic sentiment analysis were excluded. After selection, there were 24 papers. By comparing these studies with those from the previous section, we conclude that using an RNN for sentiment classification for Arabic text is still in its infancy. Figure 2 shows the number of studies on Arabic sentiment analysis using an RNN for each year. Table 3 shows the total number of papers found in each database using the mentioned keywords for Arabic text.

Fig. 2. Number of studies that use RNNs for Arabic sentiment analysis

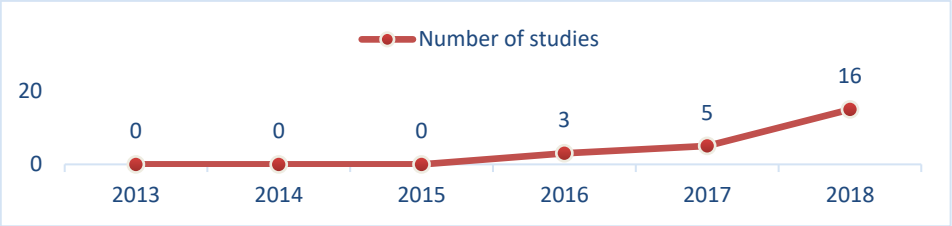

Table 3. Number of studies found by the initial search for each keyword for all databases

Databases Keyword	Springer	IEEE	Science Direct	ACM	Other
Arabic+ recurrent + neural + sentiment	38	65	37	26	807
Arabic + recurrent + neural + opinion	95	53	55	20	1,440
Total (initial search)	133	118	92	46	2,247
Total (after selection)	4	5	5	0	10

We identified 24 studies from between 2013 and 2018. These papers will be discussed in Section 5. Since the trend towards the use of RNNs for Arabic sentiment analysis in recent research is increasing, it will be an enhanced analysis of the kind of RNN used compared to the machine learning method. Although different machine learning algorithms (Adhikari et al. 2018) have been employed for sentiment analysis in the past, the sequence and latency of the meaning of the phrases of a particular text are lost. This information is retained in RNNs, which improves the prediction accuracy. In the next section, we present how SA is performed in various studies and the different techniques that have been employed.

3 Sentiment analysis approaches

In this section, we will answer RQ1. To understand recent trends in sentiment analysis using neural networks, we explain the techniques that have previously been used to obtain a sentiment analysis system and the classification levels. In Subsection 3.3, we explore word embedding (a well-known sentiment analysis technique) to extract features and represent text to be classified. In Subsection 3.4, we discuss neural networks and deep learning. Subsequently, we review related techniques that have been used for sentiment analysis, including deep learning, in Subsection 3.5.

3.1 Sentiment analysis techniques

There are two types of techniques for sentiment analysis systems (Cortes and Vapnik 1995; Madhoushi et al. 2015): machine learning and lexicon-based techniques; furthermore, there are hybrid methods that use both machine learning and lexicon-based techniques.

(i) **Machine learning techniques:** These techniques can be supervised, unsupervised, or semisupervised. They are classified into "shallow learning" approaches, such as naive Bayes (NB), maximum entropy (ME), and support vector machines (SVMs), k-nearest neighbors (KNN), and deep learning approaches using neural networks for feature learning and sentiment classification. SVMs (Cortes and Vapnik 1995) are machine learning models used for classification and regression. An SVM is a large-margin nonprobabilistic linear classifier. The principle of SVMs is to determine a hyperplane that separates training data points into two classes and keeps the margin, which is the distance between the hyperplane and the nearest data point, as large as possible,

considering the correct side of each data point. Thus, an SVM makes decisions based on the nearest data points, which are called support vectors and are chosen as the only successful elements in the training collection. Pang et al. (Pang et al. 2002) were the first to use machine learning for emotion classification. In their research, SVMs, NB, and ME were used to classify film review unigrams into two groups (positive and negative). It was demonstrated that machine learning models perform better than simple counting methods but do not perform as well in sentiment classification as traditional topic-based categorization, with an accuracy of approximately 83%.

(ii) Lexicon-based techniques: These techniques can be divided into "dictionary-based" and "corpus-based" approaches. The system does not require training to classify the data. Instead, it has predetermined sentiment values to compare the features in the text (Vohra and Teraiya 2013). A sentiment lexicon dictionary contains lists of words that express feelings and opinions; it uses scores for the opinion words to count the most dominant lexicons and determine whether a sentence is positive or negative (Liu and Zhang 2012). By contrast, a corpus relies on syntactic or co-occurrence patterns and has a large collection of texts for other opinion words. The technique starts with a list of seed opinion adjectives and a set of linguistic constraints, such as *and, but, either-or,* and *neither-nor*, to explore additional adjective opinion words and their orientations (Liu and Zhang 2012).

(iii) Hybrid techniques: The hybrid approach combines both lexicon-based and machine learning-based approaches such that the trained model considers the lexicon-based results in its features, as in (Alhumoud et al. 2015). They compared the performances of the SVM and KNN machine learning algorithms to the performance of the hybrid approach. Their experiment confirmed that the use of the hybrid approach yields better accuracy.

3.2 Neural networks and deep learning

Neural networks are complex networks with several neurons distributed in layers that simulate the function of the human brain (Schmidhuber 2015). Neural network consists of an input layer and an output layer and may include hidden layers of nonlinear processing units that link neurons. Moreover, Neural network learns features depending on the real-valued activations and suitable weights that make the neuron exhibit desired behaviors (Schmidhuber 2015).

Neural networks may be feedforward or recurrent/recursive (Sundermeyer et al. 2015). Feedforward neural networks use a straightforward data processing scheme from the input layer through a hidden layer to the output layer. In the hidden layers, there are no cycles, and thus, the output of any layer does not affect that same layer. These networks use a backpropagation algorithm to train and compute a gradient of the cost function using the most recent input to update the network parameters and thereby reduce errors during training. In contrast, recurrent/recursive neural networks contain a loop. Therefore, they can process data from prior connections/values, as well as input from the most recent layer to predict the output of the current layer. It uses temporal backpropagation, which is a regular backpropagation but calculates the gradient of a cost function using all the inputs, not just the most recent inputs.

Deep learning is a machine learning technique that uses an neural network with multiple deep layers for data processing, and thus, it learns complex features from simpler features as it proceeds from lower to higher layers using real-number activations for each neuron and weights for each link (Schmidhuber 2015).

3.3 Sentiment analysis using deep learning

In recent years, deep learning has achieved satisfactory results in natural language processing, speech recognition, and computer vision tasks (Young et al. 2018). RNNs process the sequence of the inputs and help in processing the information contained in it (Rohith et al. 2018). In sentiment analysis, several types of models using deep neural networks have been employed (Ain et al. 2017), such as CNNs (Kim 2014) and RNNs (Kobayashi et al. 2010), including bidirectional RNN (Bi-RNN) (Irsoy and Cardie 2014), long short-term memory (LSTM) (Tai et al. 2015), gated recurrent units (GRUs) (Tang et al. 2015), recursive neural networks (Socher et al. 2013), and hybrid methods. In the present study, we will focus on recurrent LSTM and GRUs. Figure 3 shows the classification techniques that have been used for the task of sentiment analysis.

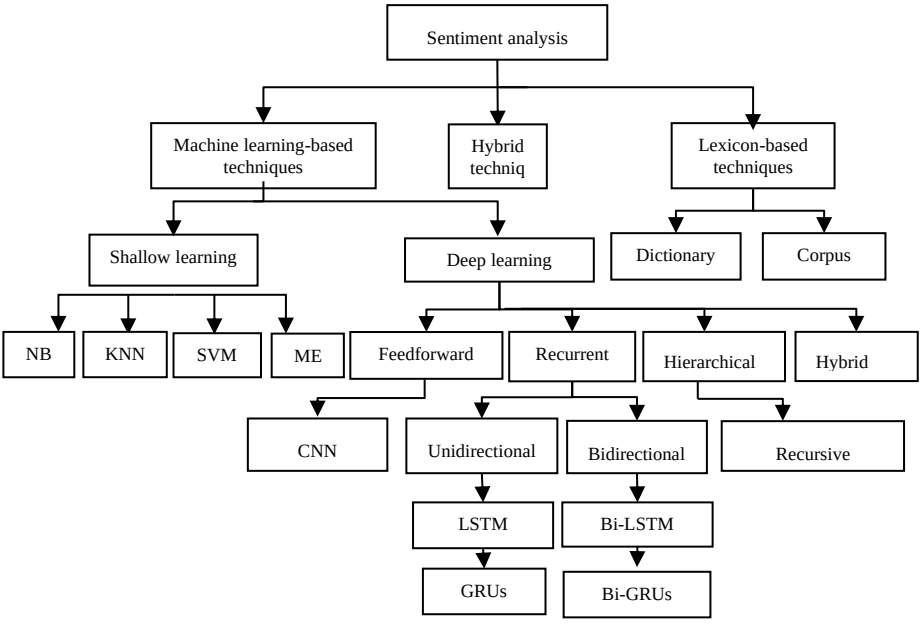

Fig. 3. Sentiment analysis techniques

One of the challenges of sentiment analysis using machine learning methods is the lack of correctly labeled data owing to difficulties related to subjective interpretation and high labor intensity. These difficulties hinder the size of the training data, affect performance, and can lower the classification accuracy (Pang and Lee 2008). To address these issues, sentiment analysis has merged with deep learning techniques owing to the automatic learning capability of the latter. This technique allows algorithms to understand sentence structure and semantics and to generate new feature representations; in contrast, traditional methods choose the most frequent word in a given input.

Given the importance of sentiment analysis using deep learning, numerous related studies involving English texts have been conducted. However, few studies have been published focusing on Arabic sentiment analysis.

3.4 Classification levels

Sentiment analysis classifications can be applied at the document, sentence, and aspect granularity levels (Zhang et al. 2018). In document-level sentiment classification, the entire

document is classified as either positive or negative. In sentence-level sentiment classification, each sentence in a document is classified as either positive, negative, or neutral. In aspect-level sentiment classification, product features are identified and extracted from the source data, where entity and aspect/feature extraction are performed along with aspect sentiment classification.

3.5 Word embedding

In sentiment analysis, words can be numerically represented before classification. Word embedding is a technique for language modeling and feature learning to enhance sentiment analysis (Zhang et al. 2018) in which features are contextually learned, and words are converted into real-valued vectors with lower dimensionality. The advantage of this technique is that words with similar meanings are represented as similar vectors. There are several available techniques for word embedding; some of the most common are briefly represented in the following. The first is manual feature extraction, such as the bag-of-words (BOW) model (Zhang et al. 2010). This is a challenging and time-consuming task, particularly for the Arabic language, which is morphologically rich. Moreover, the BOW method cannot capture relationships between words or words that have the same meanings, and bigram and trigram approaches are required to handle this issue. The second method uses Word to Vector (word2vec) (Mikolov et al. 2013), which consists of continuous BOW (CBOW) and skip-gram (SG) models. The CBOW model predicts the target word from the surrounding words within a window of a specified length. In contrast, the SG model is used to predict the surrounding words from the target word. The third method uses Global Vector (GloVe) (Pennington et al. 2014), which is trained on the nonzero entries of a global word-word co-occurrence matrix. Thus, unlike word2vec, which is a predictive model, GloVe counts the word frequency in a text.

Most existing word-embedding techniques (including those mentioned here) model only the syntactic context and ignore sentiment. However, there are some Arabic word-embedding techniques for sentiment classification (Altowayan and Tao 2016; Soliman et al. 2017; Alayba et al. 2018a).

AraVec (Soliman et al. 2017) is an example of Arabic word embedding for sentiment classification. Six different word-embedding models were constructed for the Arabic language using three different resources: Wikipedia, Twitter, and Common Crawl webpage crawl data. Two models for each resource and the SG model were provided. These models were evaluated using qualitative and quantitative measures on several tasks that involved capturing word similarity. The proposed approach presents significant results. The method can effectively identify the similarity between words and can enhance the process of other natural language processing tasks. However, character-level embedding has not been performed.

Alayba et al. (Alayba et al. 2018a) presented an Arabic word-embedding model using a 1.5-billion-word corpus. Different word2vec models were constructed using the Abu El-Khair Corpus (El-khair 2016) to choose the most suitable model for the study. A CBOW model with 200 dimensions was chosen for an automatic Arabic lexicon. It was used with different machine learning methods and CNNs and was compared with different feature selection methods for sentiment classification. In addition, a health services dataset[3] was used to test the generated model. Compared with their previous study (Alayba et al. 2017) on a health services dataset, this approach increased the sentiment classification accuracy from 0.85 to 0.92 for the main dataset and from 0.87 to 0.95 for the subdataset.

Altowayan and Tao (Altowayan and Tao 2016) used a CBOW model for word representation learning using a large Arabic corpus that includes completed texts of the Qur'an, MSA from news articles, the Arabic edition of international networks, and Dialectal Arabic from consumer reviews, with a total of 159,175 vocabulary items. To test the model,

Twitter and book review datasets for sentiment classification and news article datasets for subjectivity classification were used with six different machine-learning classification models. A comparison of the proposed subjectivity classification model with handcrafted models (Banea et al. 2010; Mourad and Darwish 2013) demonstrated that the proposed model outperforms the handcrafted models on the same dataset.

FastText[2] is another word embedding and text classification method, particularly in the case of rare words. It uses character-level information and supports 294 languages. It uses subword information in the prediction model. The representations of the words are limited, as it is difficult to represent the meaning of sentences with a single word. The techniques that identify the meanings of the sentences from words are word vectors, which are similar to autoencoders. These newer models can be used for better word embedding to identify the context of sentences.

4 RNN approaches for sentiment analysis

This section answers RQ2. To consider the latest trends in RNNs for sentiment analysis, we first explain the concept of RNNs. We will present RNNs as well as their architecture and function; moreover, we introduce different RNN architectures, such as Bi-RNN, LSTM and GRUs, which are crucial to understanding related studies that will be reviewed in the next section.

RNN architectures vary depending on the task. For example, in machine translation, several inputs are used to generate several outputs: the translated sentences. In an sentiment analysis task, several inputs are used to create a single output: the prediction, as shown in Figure 4 (Karpathy 2015).

The length of the time steps depends on the length of the input. To ensure that each comment has the same length, shorter reviews are padded with 0s in the front, based on the longest review's length K. Each word in the input is converted to embedding vectors, then these vectors are passed to RNN, LSTM or GRU layers in a recurrent manner, and each neuron shares the learned features and uses previous computations to compute the current output. Afterwards, the last prediction is chosen as the output sentiment. In the next four subsections, we present different RNN architectures: Vanilla RNN, LSTM, and GRUs. The mathematical functions based on RNN-based sentiment analysis are discussed in each subsection.

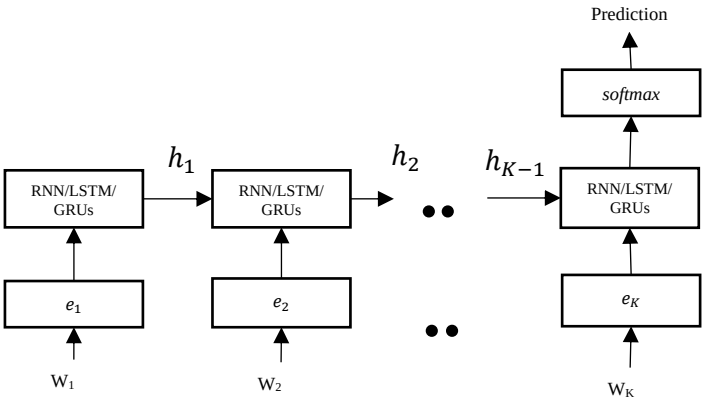

Fig. 4. Many-to-one architecture

[2] https://fasttext.cc

4.1 Vanilla RNN

In an RNN (Elman 1990; Kobayashi et al. 2010; Ayyadevara 2018), the inputs are a hidden state vector for the previous timestamp ($h_{t\ 1}$) and the input state vector at a certain time (e_t) (see Figure 4). Equation (1) generates the hidden state vector (h); W, U, and b are parameter metrics, where (W_h) is a weight matrix used to condition the input (e_t). (U_h) is a weight matrix used to condition the previous hidden state (h_{t-1}). The result of the activation function in Equation (1) is passed to the next timestamp. In the last timestep, the output is calculated as shown in Equation (2). The last output is passed to a softmax activation function to generate the predicted value $y^{\prime}$ in Equations (3) and (4), where X is a set of comments in a batch.

$$h_t = \tanh(W_h e_t + U_h h_{t\ 1} + b_h) \tag{1}$$

$$O_K = W^o h_K + b^o \tag{2}$$

$$p(y|X) = softmax\ (W^s O_K + b^s) \tag{3}$$

$$y^{\prime} = arg_y max\ p(y|X) \tag{4}$$

There are several problems with vanilla RNNs (Bengio et al. 1994). Specifically, they use an earlier sequence to make a prediction, which can lower the accuracy. This problem can be resolved using Bi-RNNs (Schuster and Paliwal 1997). Moreover, the exploding gradient problem may occur during the training process when unstable weights and directions are generated. This problem can be resolved using gradient clipping (Pascanu et al. 2013). Finally, the vanishing gradient problem, in which long-term dependencies cannot be captured, may also occur. This problem can be resolved using LSTM and GRUs.

4.2 Bi-RNN

Schuster and Paliwal first proposed Bi-RNNs (Schuster and Paliwal 1997). The principle is that the output at each timestamp depends on the output from the preceding timestamp as well as the succeeding elements in the sequence. A Bi-RNN consists of two RNNs proceeding in opposite time directions. This approach can increase the amount of input information available to the network, as information from both past and future states is used. Hence, more accurate outputs are obtained at each timestamp.

$$\overleftarrow{h_t} = \tanh(\ W_h e_t + U_h \overleftarrow{h_{t+1}} + b_h) \tag{5}$$

where the inputs are the input state vector at a certain time (e_t) and a hidden state vector for the next timestamp (h_{t+1}).

4.3 LSTM

LSTM (Hochreiter and Schmidhuber 1997; Graves et al. 2013) is an RNN that uses a complex activation function to capture long-term dependencies confidently and resolve the vanishing gradient problem using three gates. Along with the input state, it has a hidden and a cell state. In addition, the activation function and output probability are calculated at each timestamp. LSTM calculates the forget, input, and output gates using the sigmoid function σ, whose values range from 0 to 1. In Equation (6), the forget gate f is used to determine what information should be discarded from the cell state C. If the result is approximately zero, the information for the cell state will be discarded; otherwise, it will be retained. In the next step (Equation (7)), input gate I determines what new information will be added to the

cell state, as seen in. If there is no update, the value is zero. In Equation (8), a *tanh* layer generates a vector of new candidate values $\tilde{C}$.

Subsequently, in Equation (9), the cell state C is updated using Equations (6), (7), and (8). Finally, the new hidden state vector h is generated. In Equation (10), the output gate O is used to determine which parts of the cell state will be output using the sigmoid function. Subsequently, the tanh function of the new cell state C is used to obtain the hidden state in Equation (10). Equations (6), (7), (8), (9), (10) and (11) are applied in each timestep. In the last timestep, Equations (2), (3) and (4) are applied as well.

$$f_t = \sigma\,(W_f\, h_{t\ 1} + U_f\, X_t + b_f) \tag{6}$$

$$i_t = \sigma\,(W_i\, h_{t\ 1} + U_i\, X_t + b_i) \tag{7}$$

$$\tilde{C}_t = tanh\,(W_c\, h_{t\ 1} + U_c\, X_t + b_c) \tag{8}$$

$$C_t = C_{t\ 1} * f_t + i_t * \tilde{C}_t \tag{9}$$

$$O_t = \sigma\,(W_o\, h_{t\ 1} + U_o e_t + b_o) \tag{10}$$

$$h_t = O_t * tanh\,(C_t) \tag{11}$$

4.4 GRU

A GRU is a variation in LSTM. Choe et al. (Cho et al. 2014) proposed an RNN encoder–decoder and introduced the GRU model, which uses two gates (instead of three gates in LSTM) and fewer parameters; thus, it is a simpler model. The two gates are the update gate u_t, which is a combination of the forget and the input gates (Equation (12)), and the reset gate *rt*, which indicates the relevance of the previous cell state for computing the next candidate (Equation (13)). Initially, the cell state equals the hidden state; that is, a *tanh* layer generates a vector of new candidate values $\tilde{C}$ in Equation (14) using the reset gate *rt*. Subsequently, in Equation (15), the hidden state *h* is updated using Equations (11) and (13). Equations (12), (13), (14) and (15) are applied in each timestep. In the last timestep, Equations (2), (3) and (4) are applied as well.

$$u_t = \sigma\,(W_u\, h_{t\ 1} + U_u\, X_t + b_u) \tag{12}$$

$$r_t = \sigma\,(W_r\, h_{t\ 1} + U_r\, X_t + b_r) \tag{13}$$

$$\tilde{C}_t = tanh\,(W_c . [r_t * h_{t\ 1}] + U_c\, e_t + b_c) \tag{14}$$

$$h_t = u_t * \tilde{C}_t + (1 - u_t) * h_{t\ 1} \tag{15}$$

According to Chung et al. (Chung et al. 2014), there are similarities and differences between LSTM and GRUs. The most prominent common feature is their ability to retain existing components and add the updated component instead of replacing the entire activation function as in conventional recurrent units. Accordingly, long-term dependencies may be captured, and shortcut paths may be created so that the error can be easily backpropagated without rapidly vanishing, thus eliminating the vanishing gradient problem.

A difference between LSTM and GRUs is that the former controls the memory content and cell state exposure to other units using the output gate, whereas the latter exposes the entire cell state to other units without control. Another difference, as mentioned earlier, is that the LSTM unit has separate input and forget gates, whereas the GRU merges these operations through the reset gate. The LSTM unit does not control the information flow from the previous timestamp, but it controls the amount of the new content being added to

the memory cell from the forget gate. By contrast, a GRU controls the information flow from the previous activation when computing the new candidate activation but does not control the amount of the candidate activation being added, as the control is performed through the update gate.

5 Related works for Arabic sentiment analysis using RNNs

This section addresses RQ3 by discussing related studies that use RNN models for Arabic sentiment analysis. There are only 24 such studies. The studies under consideration are those that introduce a new Arabic dataset for sentiment analysis and use an RNN to test the accuracy of these corpora. Moreover, as Arabic datasets are scarce and limited, we highlight the various datasets used in all studies under consideration in Table 4.

The studies are divided according to the classification level (sentence or aspect) and the type of analysis (emotion detection, emoji analysis, hate speech detection, or sentiment classification).

We discuss the 6 papers related to aspect-based sentiment analysis in Section 5.1. In addition, we discuss the 18 papers related to sentence level in Section 5.2; 5 papers related to emotion detection (Section 5.2.1), 3 papers on emoji analysis (Section 5.2.2), 1 paper related to hate speech detection (Section 5.2.3), and 9 papers focused on only sentiment classification (Section 5.2.4).

Table 4. Related studies that used Arabic datasets for sentiment analysis using an RNN

Datasets	Size	Classes	Arabic form	Papers
Large-scale Arabic book reviews dataset (LABR) (Aly and Atiya 2013).	63,257 book reviews.	3 classes	MSA/Dialectal.	(Abbes et al. 2017). (Baniata and Park 2016).
SemEval-2016 Task 5 (Pontiki et al.).	~3,000 hotel reviews.	4 classes	MSA/Dialectal.	(Tamchyna and Veselovská 2016). (Ruder et al. 2016a). (Wang and Lu 2018). (Ponti et al. 2017). (Al-Smadi et al. 2018). (Al-Smadi et al. 2019).
Arabic Sentiment Tweets Dataset (ASTD) (Nabil et al. 2015).	10,006 tweets.	4 classes	Egyptian dialect.	(Al-Azani and El-Alfy 2017). (Al-Azani and El-Alfy 2018). (Baccouche et al. 2018). (Alayba et al. 2018b). (Heikal et al. 2018).
ArTwitter (Abdulla et al. 2013).	2,000 tweets.	2 classes	MSA/Jordanian dialect.	(Al-Azani and El-Alfy 2017). (Alayba et al. 2018b). (Al-Azani and El-Alfy 2018).
Qatar Computing Research Institute dataset (QCRI) (Mourad and Darwish 2013).	2,300 tweets.	(objective, subjective), 2 classes	MSA/Egyptian, Levantine, and Gulf dialects.	(Al-Azani and El-Alfy 2018).
Syria (Mohammad et al. 2016).	2,000 tweets.	2 classes, 3 classes	Syrian dialect.	(Al-Azani and El-Alfy 2018).
SemEval-2017 Task 4 (Rosenthal et al. 2017).	1,656 (2 classes) + 3,355 tweets.	2 classes, 3 classes 5 classes	Dialectal.	(Al-Azani and El-Alfy 2018). (González et al. 2017). (Samy et al. 2018).
SemEval-2018 Task 1 (Mohammad et al. 2018).	11,781 tweets.	7 classes, 12 emotion classes	Dialectal.	(Abdullah and Shaikh 2018). (Abdullah et al. 2018). (Samy et al. 2018). (Alhuzali et al. 2018). (Abdou et al. 2018).
DINA (Abdul-Mageed et al. 2016).	3,000 tweets.	6 emotion classes	Dialectal.	(Alhuzali et al. 2018).
AraSenTi (Al-Twairesh et al. 2017).	17,573 tweets.	2 classes	Saudi dialect.	(Alwehaibi and Roy 2018).
MD-ArSenTD	14,400 tweets.	3 classes	Dialectal.	(Baly et al. 2017b).
BRAD 2.0	692,586 book reviews.	3 classes	MSA/Dialectal.	(Elnagar et al. 2018).
Arabic Health Services Dataset	2,026 tweets.	2 classes	Dialectal.	(Alayba et al. 2018b).
Collected tweets.	12,897 tweets.	2 classes	Dialectal	(El-Kilany et al. 2018).
	6,600 tweets.	2 hate speech classes	Dialectal.	(Albadi et al. 2018).
ArSenL	157,969 synsets	3 classes	MSA	(Badaro et al. 2014).
Semeval-2016 Task 7	1,366 tweets	4 classes	MSA/Dialectal	(Kiritchenko et al. 2016).
Arabic Gold Standard Twitter Data	4,191 tweets	3 classes	MSA/Dialectal	(Refaee and Rieser 2014)
ArSAS	19,762 tweets	3 classes	MSA/Dialectal	(Elmadany et al. 2018)

5.1 Aspect-based sentiment analysis using an RNN

In this section, we focus on studies that classify text at the aspect level. These works are categorized based on RNN variations (RNN, LSTM, and Bi-LSTM).

Al-Smadi et al. (Al-Smadi et al. 2018) proposed an aspect-based sentiment analysis of Arabic hotel reviews using two implementations: deep RNN and SVM, along with word embedding, lexical, syntactic, morphological, and semantic features. The F1 score was employed to evaluate aspect opinion target expression extraction (T1) and aspect category identification (T2) and the accuracy score to evaluate sentiment polarity identification (T3). Different models were used for performance comparison: a baseline model (AL-Smadi et al. 2016), INSIGHT-1 (Ruder et al. 2016b), and UFAL (Tamchyna and Veselovská 2016). The results demonstrated that the SVM outperformed the RNN approach in all tasks because the feature sets used to train the SVM were large, particularly in T3, with an accuracy of 95.4%, compared with RNN with an accuracy of 87%. However, the deep RNN outperformed the SVM in terms of the execution time, particularly in T2, with a speed rate of 5.3× compared with T1, with a speed rate of 4.4×, and T3, with a speed rate of 2.1×.

Tamchyna and Veselovska (Tamchyna and Veselovská 2016) employed a multilingual LSTM model for aspect-based sentiment analysis using the categories of consumer electronics, restaurants, and hotel reviews in seven languages: Arabic, Dutch, English, French, Russian, Spanish, and Turkish. The main goal was to discover the linguistic patterns in the data automatically without feature extraction or language-specific tools. The model was compared with five different versions of the same model: the official baseline model (provided by the task organizer) using an SVM; a baseline model using logistic regression without any type of feature except BOW, a model submitted for official evaluation, where the networks were not fully optimized by the submission deadline; an optimized model, where the results were obtained after the deadline; and the best model, where the best score for each language and domain was reported. Regarding the Arabic language aspect-based sentiment analysis, the optimized model exhibited the best results in terms of the F1 score (52.59%) compared with all the other models, particularly the best model, which achieved an F1 score of 52.11%.

Al-Smadi et al. (Al-Smadi et al. 2019) improved the deep learning approach in (Al-Smadi et al. 2018) for aspect-based sentiment analysis of Arabic hotel reviews using LSTM: first, with a character-level bidirectional LSTM along with a conditional random field (CRF) classifier for aspect opinion target expression extraction, and second, with an aspect-based LSTM for aspect sentiment polarity classification in which the aspect opinion target expressions are considered attention expressions to support sentiment polarity identification. For sentiment polarity identification, the authors compared the proposed model with the baseline model (AL-Smadi et al. 2016), INSIGHT-1 (Ruder et al. 2016b) and IIT-TUDA (Kumar et al. 2016). The authors used the F1 score to evaluate opinion target expression extraction and the accuracy to evaluate sentiment polarity identification. The results demonstrated that this approach outperformed the baseline on both tasks, with an improvement of 39% for aspect opinion target expression extraction and 6% for aspect sentiment polarity classification. For opinion target expression extraction, the proposed model yielded promising results, particularly when FastText was used for character embedding, with an F1 score of 69.98%, compared with the same model when word2vec was used, which achieved an F1 score of 66.32%. Moreover, for sentiment polarity identification, the proposed model outperformed other models, with an accuracy of 82.6%, except INSIGHT-1, which used a CNN and had an accuracy of 82.7%.

Ruder et al. (Ruder et al. 2016a) presented a hierarchical bidirectional LSTM that is able to consider intrasentence relations such as background and intersentence relations (e.g., aspect-based sentiment analysis using bidirectional sentence-level LSTM and bidirectional review-level LSTM). The evaluation was performed using seven different models: the best

model for each domain and language (Pontiki et al. 2016), XRCE (Brun et al. 2016), IIT-TUDA (Kumar et al. 2016), a sentence-level CNN (INSIGHT-1) (Ruder et al. 2016b), a sentence-level LSTM, which is the first layer of the proposed model, and the proposed model with randomly initialized word embeddings (H-LSTM) and with pretrained embeddings (HP-LSTM). Regarding the Arabic language analysis, the proposed model with pretrained word embeddings outperformed the other models, including the sentence-level models, with an accuracy of 82.9% compared with the best model, which achieved an accuracy of 82.7%.

Wang and Lu (Wang and Lu 2018) proposed a segmentation attention-based Bi-LSTM model for aspect-based sentiment analysis that extracts sentiment information and learns latent opinions by capturing the structural dependencies between the given target and the sentiment expressions using a CRF layer. The proposed model was tested on two groups of datasets. One group was from online reviews (laptops and restaurants) and social comments on Twitter, and SemEval 2014 Task 4 was used to analyze each component of the model. In the other group, the SemEval 2016 Task 5 dataset (Pontiki et al.) was used to examine the model's language sensitivity. The evaluation was performed by comparing the extracted opinions with the manually annotated opinions using several models from Ruder et al. (Ruder et al. 2016a), LSTM with standard attention *softmax* (A-LSTM), and the proposed model with segmentation attention layer with and without penalty terms (SA-LSTM and SA-LSTM-P, respectively). Regarding the hotel reviews in Arabic, the experiments demonstrated that the proposed model with penalty achieved the best results (an accuracy of 86.9%) compared with other models, particularly with the proposed model without penalty, which achieved an accuracy of 86.7%.

Ponti et al. (Ponti et al. 2017) examined the amount of information that representations retain about the polarity of sentences for each language. Moreover, they presented a model that decodes sentiment from unsupervised sentence representations learned by different architectures (sensitive to words, sensitive to order, or neither), such as additive skip-gram, FastSent, sequential denoising autoencoder, and distributed BOW (DBOW) models, and they compared these models with bidirectional LSTM. Regarding the Arabic language sentiment analysis, the DBOW model yielded the best results compared with other unsupervised representations, with a weighted F1 score of 76.76%, whereas the SDAE model achieved an F1 score of 72.13%. However, Bi-LSTM outperformed all the unsupervised strategies, with an F1 score of 86.56%. Bi-LSTM uses the skip-gram model for sentence representation; hence, an RNN model can be chosen as a ceiling, especially for the Arabic language. Table 5 shows the results for aspect-based sentiment analysis for Arabic text using RNNs.

Table 5. Aspect-based sentiment analysis for Arabic text using RNN

Papers	Classifier	Word embedding	Dataset	Accuracy
(Al-Smadi et al. 2018).	RNN.	Word2vec.	SemEval-2016 Task 5 Arabic Hotel reviews (Pontiki et al.).	87%.
(Tamchyna and Veselovská 2016).	LSTM.	Word2vec: CBOW.	SemEval-2016 Task 5. including Arabic Hotel reviews (Pontiki et al.).	F1 score: 52.59%.
(Al-Smadi et al. 2019).	LSTM.	Word2vec and FastText for character-level embedding.	SemEval-2016 Task 5 Arabic Hotel reviews (Pontiki et al.).	LSTM: 82.6%
(Ruder et al. 2016a).	Hierarchical Bi-LSTMs.	Trained embedding by Leipzig Corpora Collection.	SemEval-2016 Task 5, including Arabic Hotel reviews (Pontiki et al.).	LSTM: 80.5%. H-LSTM: 82.8%. HP-LSTM: 82.9%.
(Wang and Lu 2018).	Bi-LSTM + CRF	Trained embedding by Leipzig Corpora Collection.	SemEval-2016 Task 5, including Arabic Hotel reviews (Pontiki et al.).	A-LSTM: 86.5%. SA-LSTM: 86.7%. SA-LSTM-P: 86.9%.
(Ponti et al. 2017).	Bi-LSTM.	Additive skip-gram and ParagraphVec DBOW. FastSent, SDAE.	SemEval-2016 Task 5, including Arabic Hotel reviews (Pontiki et al.).	Weighted F1 score: 86.56%.

It should be noted that no study used GRUs, Bi-GRUs, or hybrid methods for aspect-based classification. In addition, multilingual models, as shown in (Tamchyna and Veselovská 2016), were not effective in improving the Arabic language sentiment analysis specifically. Moreover, all the studies depended on a unified dataset, demonstrating the need for well-annotated datasets for aspect terms, categories, and sentiment polarity.

5.2 Sentence-level sentiment analysis using an RNN

In this part, we will divide the studies according to the type of analysis involved, namely, emotion detection, emoji analysis, hate speech detection, and sentiment classification.

5.2.1 Emotion detection

This section will introduce studies concerned with the detection and classification of emotions such as anger, fear, joy, and sadness.

Samy et al. (Samy et al. 2018) proposed using GRU and context-aware GRU (C-GRU) architectures to investigate the role of social influence on shaping others' opinions and emotions in the same environment and the effect on the determination of the sentiment polarity. These models extracted contextual information (topics) and used both topic and sentence information to detect multilabel emotions. The C-GRU architecture was compared a support vector classifier with a linear kernel, L1 regularization from (Badaro et al. 2018), and a context-free GRU architecture. The results demonstrated that the C-GRU architecture achieved an accuracy of 53.2%, a macro-average F1 of 64.8%, and a micro-average F1 of 49.5%, outperforming the simple GRU architecture, which achieved an accuracy of 52.4%, a macro-average F1 of 64.2%, and a micro-average F1 of 49.6%.

Abdullah and Shaikh (Abdullah and Shaikh 2018) presented TeamUNCC's system, which used an LSTM network to detect emotion intensity or sentiment in English, Arabic, and translated Arabic tweets from SemEval 2018. The system attempted to complete all five subtasks and determine the intensity and sentiment of the tweets from SemEval 2018. The tweets were preprocessed and fed into word2vec, doc2vec, and other feature vectors for feature extraction. Subsequently, these vectors were fed into the deep neural network layers

for prediction. The Spearman correlation scores demonstrated that the proposed system yielded promising results, with an emotion regression score of 59.7%, an emotion classification score of 51.7%, a sentiment regression score of 77.3%, and a sentiment classification score of 74.8%, compared with the baseline model by the SemEval Task 1 organizers, which is based on SVM-Unigrams and achieved an emotion regression of score 45.5%, an emotion classification score of 31.5%, a sentiment regression score of 57.1%, and a sentiment classification score of 47.1%.

Abdullah et al. (Abdullah et al. 2018) presented a new version of TeamUNCC's system, called SEDAT, to explore the emotional intensity and sentiment for Arabic tweets. This model consisted of two submodels: the first used a collection of Arabic tweets with five dimensions and translated tweets with 4,903 dimensions using a set of features to produce vectors, and the other used only Arabic tweets with 300 dimensions, which are trained using skip-gram-Twitter from AraVec to produce vectors. The first submodel passed generated vectors to a fully connected neural network, whereas in the second submodel, the vectors were fed into a CNN-LSTM model. The output layer consisted of one sigmoid neuron that produced a real-valued number between 0 and 1. The Spearman correlation scores demonstrated that SEDAT outperformed TeamUNCC's system (Abdullah and Shaikh 2018) with an emotion regression score of 66.1%, an emotion classification score of 56.9%, a sentiment regression score of 81.7%, and a sentiment classification score of 78.6%; moreover, SEDAT is only 1 to 2 points behind the state-of-the-art models, that is, AffecThor for emotion (58.7%) and EiTAKA for sentiment (80.9%) (Mohammad et al. 2018).

Abdou et al. (Abdou et al. 2018) presented AffecThor, which consists of three different models. These models used learned and manually crafted representations, such as character embedding, word embedding, inference, and average lexicon representations. The proposed models used a feedforward neural network or gradient boosted trees for regression and two ensemble regressors: simple averaging cross-validation and a nonlinearity (sigmoid) layer on top of the different submodels, such as CNN and Bi-LSTM. The use of simple averaging provides the best results for all the SemEval 2018 models in the Arabic language, particularly in emotion classification, with an emotion classification score of 58.7%.

Alhuzali et al. (Alhuzali et al. 2018) described and comprehensively confirmed a technique for the mechanical reproduction of labeled emotion information; an improved dataset was also used to extract emotions from modern and dialectical Arabic text, which focused on Robert Plutchik's eight core emotions. Using a mixed supervision method that exploits the seeds of first-person feelings, it was also demonstrated that promising results can be obtained through an RNN with deep gates. Alhuzali et al. extended the manually annotated dataset DINA (Abdul-Mageed et al. 2016) to LAMA. The proposed dataset was based on emotion existence and intensity in one stage for English and Arabic. The proposed approach is based on emotion phrase seeds from Robert Plutchik's eight basic emotion types: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust. In addition, the authors proposed a hybrid supervised method that automatically determines emotion intensity and sentiments for English and Arabic, and they compared it with supervised and distant supervised methods. The proposed method was applied to baseline models, such as a multinomial NB classifier, a passive-aggressive classifier, a perceptron classifier (PTN), an SVM classifier, an SVM classifier trained with stochastic gradient descent (SVM-SGD), and the proposed GRU classifier. In addition, the models were validated using SemEval-2018 (Mohammad et al. 2018). The results demonstrated that the GRU model yielded the best results, particularly in terms of emotion detection, on all datasets, particularly the DINA and LAMA-DINA datasets, with an F1 score of 98%, compared with SVM-SGD, which achieved an F1 score of 92%. Moreover, a hybrid supervision of the LAMA-DINA and LAMA-DIST datasets using the GRU model yielded the best results in terms of emotion classification, with an average F1 score of 70% compared with SVM-SGD, which achieved an F1 score of 60%. Table 6 shows the emotion detection models for Arabic text.

Table 6. Sentence/document/word-level sentiment analysis based on emotion detection for Arabic text

Papers	Classifier	Sentiment Classification Level	Document/Text Representation	Dataset	Accuracy
(Samy et al. 2018).	GRU, C-GRU.	Sentence level.	AraVec (Soliman et al. 2017).	From SemEval-2017 (Rosenthal et al. 2017) and SemEval-2018 (Mohammad et al. 2018).	GRU: 53.2%. C-GRU: 52.4%.
(Abdullah and Shaikh 2018).	Dense and LSTM networks.	Sentence level and document level.	Word2vec, doc2vec with different feature vectors.	From SemEval-2018 (Task 1: Affect in Tweets) (Mohammad et al. 2018): raw Arabic and Arabic translated into English.	Emotion detection: 44.6%. Spearman correlation scores: emotion classification 51.7%. sentiment classification 74.8%.
(Abdullah et al. 2018).	CNN-LSTM	Sentence level and document level.	AraVec(Soliman et al. 2017), doc2vec and a set of semantic features.	From SemEval-2018 (Task 1: Affect in Tweets) (Mohammad et al. 2018): raw Arabic and Arabic translated into English.	Spearman correlation scores: emotion classification 56.9%. sentiment classification 78.6%.
(Abdou et al. 2018).	CNN, Bi-LSTM	Sentence level and word level.	Word2vec skip-gram embeddings.	From SemEval-2018 (Task 1: Affect in Tweets) (Mohammad et al. 2018).	Spearman correlation scores: Emotion classification 58.7%. Sentiment classification 75.2%.
(Alhuzali et al. 2018).	GRU.	Sentence level.	Word2vec.	LAMA, DINA (Abdul-Mageed et al. 2016), DIST and SemEval-2018 (Mohammad et al. 2018).	LAMA-DINA+LAMA-DIST: F1 score 70%.

As shown in Table 6, there is a lack of research on emotion detection and analysis using neural networks. Moreover, it is noted that there is a dominant dataset that is used for emotion analysis studies, namely, SemEval-2018 (Task 1: Affect in Tweets).

5.2.2 Emoji analysis

This section introduces studies on emoji analysis using RNNs. Emoji are ideograms and smileys that are used frequently on social media to express ideas and emotions. They differ from emoticons, which use letters, numbers and symbols to create icons.

Al-Azani and El-Alfy (Al-Azani and El-Alfy 2018) used deep RNN models, namely, LSTM, Bi-LSTM, GRU, and Bi-GRU, with different modes (summation, multiplication, concatenation, and average of outputs) for emoji-based tweets to detect sentiment polarity and compared these models with deep neural networks and baseline machine learning classifiers, such as stochastic gradient descent, Gaussian NB, SVM, KNN, and decision tree classifiers. These models used a set of 843 Arabic microblogs with emojis from different resources, such as the ASTD (Nabil et al. 2015), ArTwitter (Abdulla et al. 2013), QCRI (Mourad and Darwish 2013), Syria (Mohammad et al. 2016), and SemEval-2017 Task 4 Subtask A (Rosenthal et al. 2017); in addition, data were collected from Twitter and YouTube and were manually annotated. They used the Emoji Sentiment Ranking lexicon to

identify emojis in the dataset. The performance was evaluated in terms of the precision, recall, F1 score, accuracy, Matthews correlation coefficient, and geometric mean. The results demonstrated that the LSTM and GRU models significantly outperformed the other models. Specifically, the bidirectional GRU performed best, with an accuracy of 78.71% and an F1 score of 78.76%, compared with Bi-LSTM, which had an accuracy of 77.99% and an F1 score of 78.10%.

Baccouche et al. (Baccouche et al. 2018) proposed an automatic labeling technique for health-related tweets in three different languages: English, French, and Arabic. In addition, they applied sentiment analysis models such as a CNN model and an LSTM model to classify tweets into two and three classes. The dataset contained both a domain-specific health-related dataset and a nonspecific domain dataset from Amazon, IMDB, Yelp, and ASTD. The automatic labeling technique preprocessed the health-related tweets to detect the annotations using domain knowledge, natural language processing, and sentiment lexicon dictionaries. The proposed annotation models yielded the best results using an RNN, particularly for English, with an accuracy of 98% and an F1 score of 97%, compared with the CNN, which had an accuracy of 97% and an F1 score of 96%. Moreover, the proposed model improved by 1.1% when a nonspecific domain dataset was added to the LSTM model.

Heikal et al. (Heikal et al. 2018) presented a model that combined CNN and Bi-LSTM to predict the sentiment of Arabic tweets using the ASTD dataset. This model followed the same method as in a previous study for English (Cliche 2017), using various hyperparameters to improve the accuracy and handling emoticons by mapping them to the related emojis. Although the proposed model did not use any feature engineering to extract special features, it achieved the best results, with an accuracy of 65.05% and an F1 score of 64.46% compared with the state-of-the-art model from Baly et al. (Baly et al. 2017a), which achieved an accuracy of 58.5% and an F1 score of 53.6%. Table 7 summarizes the studies that focus on emoji analysis for Arabic text.

Table 7. Emoji analysis for Arabic text

Papers	Classifier	Document/Text Representation	Dataset	Accuracy
(Al-Azani and El-Alfy 2018).	LSTM, Bi-LSTM, GRU, Bi-GRU.	Emoji Sentiment Ranking lexicon.	Combined datasets from ASTD (Nabil et al. 2015), ArTwitter(Abdulla et al. 2013), QCRI (Mourad and Darwish 2013), Syria (Mohammad et al. 2016), Semeval-2017 Task 4 Subtask A(Rosenthal et al. 2017) and other resources	Bi-GRU: 78.71%. Bi-LSTM: 77.99%.
(Baccouche et al. 2018).	LSTM.	Word2vec. (pretrained by Wikipedia).	Health-related dataset (the authors did not mention the final number) and a non-health-related dataset compiled from Amazon, IMDB, Yelp and the ASTD (Nabil et al. 2015).	83%.
(Heikal et al. 2018).	CNN-Bi-LSTM.	AraVec (Soliman et al. 2017).	ASTD dataset (Nabil et al. 2015).	CNN-LSTM: 65.05%.

Only three studies performed emoji analysis; the best accuracy was 83%, which highlights the need for more research in this direction. As emojis are used in a paradoxical way, a robust model that detects sarcasm and actual sentiment is required. In addition, there is a scarcity of datasets containing emojis. Accordingly, more datasets need to be created along with emoji detection algorithms.

5.2.3 Hate speech detection

Hate speech is a type of public speech that encourages hate or violence towards a person or a community on the basis of a trait. Only one study (Albadi et al. 2018) focused on hate speech detection in Arabic. This is a fairly new topic, and the study introduced the first dataset that can be used to address this issue.

Albadi et al. (Albadi et al. 2018) provided the first dataset for detecting religious hate speech in Arabic tweets. It consists of 6,000 labeled tweets. Moreover, they created the first three Arabic lexicons consisting of common terms in religious discussions, with scores that describe the polarity and strength of these terms. In addition, they developed three different approaches to capture religious hate speech: a lexicon-based approach using three Arabic lexicons, an n-gram-based approach using logistic regression and SVM, and a deep learning-based approach using GRU with AraVec embedding (Soliman et al. 2017). The GRU model achieved the best results compared with all the other models, with an accuracy of 79%, an F1 score of 77%, and an area under the receiver operating characteristic curve of 84%. The second-best model (SVM) had an accuracy of 75%, an F1 score of 72%, and an area under the receiver operating characteristic curve of 81%.

Detecting hate speech is an important feature of controlling spam and arguments in public forums. Hence, more studies must be conducted to analyze and detect hate speech. Additionally, the existing method has a low accuracy of only 79%, which cannot be considered significant. Therefore, higher accuracies must be targeted to effectively improve the detection of hate speech.

5.2.4 Sentiment classification

In this section, we divide the related studies into two parts: first, studies that used LSTM, GRU, or both, and second, studies that used hybrid models, where an additional neural network was used as a layer in the model along with an RNN such as a CNN. A summary is presented in the following.

LSTM/GRU models. Abbes et al. (Abbes et al. 2017) proposed two models for deep learning-based sentiment analysis: a deep neural networks and an RNN for Arabic social media. The proposed models followed a number of steps. The first stage was gathering and collecting data from the LABR (Aly and Atiya 2013). The second stage was preprocessing and building a lexicon model. The third phase was the feature extraction step, consisting of deriving the lexicon-based relevant features from the stored data using word embedding. The final steps involved applying deep learning models, that is, deep neural networks and RNN (LSTM) to classify sentence polarity. The experimental results demonstrated that the RNN outperformed the deep neural network, with an accuracy of 71%, a precision of 68.3%, a recall of 77%, and an F1 score of 72.4%, whereas the deep neural networks achieved an accuracy of 64.4%, a precision of 61.1%, a recall of 75.3%, and an F1 score of 67.5%. Although the comparison was made using 300 epochs for the deep neural networks and only 30 epochs for the RNN was unfair, the highest result was achieved at epoch 200 for the deep neural networks and epoch 12 for the RNN.

Alwehaibi and Roy (Alwehaibi and Roy 2018) used LSTM to classify Arabic texts with different pretrained word-embedding techniques, namely, AraVec, ArabicNews, and Arabic FastText, to investigate the effect of these techniques on the model accuracy. Initially, the model preprocessed the AraSenTi tweet datasets. Subsequently, the tweets were processed by a pretrained word-embedding technique to generate vectors for each word. Then, the embedding was fed to the LSTM layer with a 128-dimensional hidden state to classify each tweet as positive, negative, or neutral. Arabic FastText achieved the best accuracy (93.5%) compared with AraVec (88%) and ArabicNews (91%). While the dataset is divided equally into three classes, the low F1 score (43% for ArabicNews, 40% for AraVec, and 41% for

Arabic FastText) is an indication of both poor precision and poor recall.

Baly et al. (Baly et al. 2017b) provided the first multidialectal Arabic sentiment Twitter dataset (MD-ArSenTD). It contains annotated tweets (for both sentiment and dialect) that were collected from 12 Arab countries from the Gulf, Levant, and North Africa. In addition, the authors analyzed tweets from Egypt and the United Arab Emirates (UAE) to investigate regional variations in the data characteristics and their effect on sentiment by using feature engineering for an SVM model and generic and dialect-specific embeddings for an LSTM model. The results demonstrated the superiority of the LSTM model over the SVM model, particularly when lemma embedding was used for Egyptian tweets, with an accuracy of 70.0% and a weighted F1 score of 69.1%, compared with UAE tweets, with an accuracy of 63.7% and a weighted F1 of 64.8% for the 3-class classification tasks.

Elnager et al. (Elnagar et al. 2018) presented book reviews in the Arabic dataset BRAD 2.0, which is an extension of BRAD 1.0 with more than 200,000 additional records and a total of 692,586 annotated reviews. This dataset combines both MSA and dialectical Arabic and three classes of reviews: positive, negative and neutral. In the experiment, only two classes were used. The authors applied NB, decision tree, random forest, XGBoost, SVM, CNN and LSTM classifiers to verify and validate the proposed dataset. The machine learning classifiers generally performed better using unigrams and bigrams, especially without TF-IDF, while the SVM classifier achieved the best result with TF-IDF with an accuracy of 90.86%. For the deep learning models, the results show that the LSTM classifier outperformed the CNN classifier, especially when using GloVe, with an accuracy of 90.05% vs. 90.02%, respectively. However, an imbalanced dataset led to bias for the positive class: 64% of the data were classified as positive, 15% as negative, and 21% as neutral.

El-Kilany et al. (El-Kilany et al. 2018) presented a model that recognized the sentiment targets from Arabic tweets using two layers: a character and word2vec embedding layer and a bidirectional LSTM with a CRF classification layer. The Bi-LSTM used vectors produced by the first layer to generate the features for the CRF classifier, which used these contextual features to predict a tag for the current word based on the previous and subsequent word tags in the sentence. The results demonstrated that the proposed model achieved higher recall (71.4%) than the same model that used only character embedding without word2vec, which achieved a recall of 61.5%. This finding implies that the proposed model discovered more target entities from the tweets and maintained almost the same extraction precision (approximately 73%).

Table 8 summarizes the sentence/character-level sentiment analysis using LSTM/GRU for Arabic text.

Table 8. Sentence-level/character-level sentiment analysis using LSTM/GRU for Arabic text

Papers	Classifier	Classification Level	Document/Text Representation	Dataset	Accuracy
(Abbes et al. 2017).	LSTM.	Sentence level.	TF-IDF, BOW.	LABR (Aly and Atiya 2013).	64.4%.
(Alwehaibi and Roy 2018).	LSTM.	Sentence level.	AraVec ArabicNews, and Arabic FastText.	AraSenTi dataset (Al-Twairesh et al. 2017).	AraFT: 93.5%. AraVec: 88%. ArabicNews: 91%.
(Baly et al. 2017b).	LSTM.	Sentence level.	Word2vec, lemma embedding, and stem embedding: skip-gram.	MD-ArSenTD.	Egyptian tweets: 70.0% UAE tweets: 63.7%.
(Elnagar et al. 2018).	LSTM.	Sentence level.	GloVe (Pennington et al. 2014), TF-IDF.	BRAD 2.0 (Elnagar et al. 2018).	90.05%.
(El-Kilany et al. 2018).	Bi-LSTM + CRF.	Sentence and character level.	Word2vec and character embedding.	3,000 tweets	F1 score: 72.6%.

Hybrid models. Al-Azani and El-Alfy (Al-Azani and El-Alfy 2017) compared various CNN and LSTM approaches for sentiment analysis of Arabic microblogs using six models: LSTM, CNN, CNN with LSTM, three-stacked LSTM layers, and two LSTMs combined with summation, multiplication, and concatenation. These models were evaluated using four evaluation measures: the precision, recall, accuracy, and F1 score. Two benchmark Arabic tweet datasets were used: ASTD (Nabil et al. 2015) and Arabic sentiment analysis ArTwitter (Abdulla et al. 2013). Word2vec was used as the input of the investigated models, with static and nonstatic word initialization for CBOW and skip-gram word embedding. The experiments demonstrated that using word2vec vectors updated during learning achieved the best results in nearly all cases. In addition, the LSTM model outperformed the CNN model. Moreover, nonstatic models with the combined LSTM architectures outperformed the other models, particularly when two LSTMs were combined with concatenation for the ArTwitter dataset and with skip-gram word embedding. The precision, recall, accuracy, and F1 score reached 87.36%, 87.27%, 87.27%, and 87.28%, respectively. By contrast, the same architecture for the ArTwitter dataset and with CBOW word embedding achieved a precision of 86.46%, a recall of 86.45%, an accuracy of 86.45%, and an F1 score of 86.45%.

Alayba et al. (Alayba et al. 2018b) presented a model that combined a CNN and LSTM for sentiment classification at the character level, character 5-gram level, and word level to expand the number of features, and they investigated its accuracy using different datasets, such as the main Arabic health services dataset and a subset (Alayba et al. 2017), ArTwitter (Abdulla et al. 2013), and ASTD (Nabil et al. 2015). Moreover, the proposed model was compared with those by Alayba et al. (Alayba et al. 2018a), Abdulla et al. (Abdulla et al. 2013), and Dahou et al. (Dahou et al. 2016). Although the model by Dahou et al. achieved the best results for the ASTD dataset, with an accuracy of 79.07%, the proposed model achieved promising accuracy, particularly at the character 5-gram level and with the subset of the Arabic health services dataset (95.68%), compared with the word-level model on the main Arabic health services dataset (accuracy of 94.24%). These high accuracies are due to the imbalanced datasets for the main Arabic health services dataset and the subset. Although the character, character 5-gram, and word-level models reached accuracies of 74.19%, 77.62%, and 76.41%, respectively, for the ASTD dataset, the proposed model from (Al-Azani and El-Alfy 2017) showed the best results with LSTM-MUL and an accuracy of 81.63% with CBOW, which shows the importance of word-embedding techniques over row-embedding techniques.

Gonzalez et al. (González et al. 2017) proposed the ELiRF-UPV system for enhancing performance on Task 4 of SemEval-2017. It involved five different subtasks: (A) message polarity classification, (B) and (C) tweet classification according to a two- and five-point scale, respectively, and (D) and (E) tweet quantification according to a two- and five-point scale, respectively. The proposed model used a CNN and an RNN and the combination of general and specific word embeddings with polarity lexicons using both English and Arabic. The system used three CRNNs (a CNN, max pooling, and Bi-LSTM) to extract spatial relations among the words of a sentence with an input of three different embeddings: out-domain embeddings, in-domain embeddings, and sequences of word polarities. The outputs of these networks were used as the input of a fully connected multilayer perceptron. Different measurements were made to evaluate each task. Regarding subtask A, the proposed model using Arabic achieved an accuracy of 50.8%, whereas the proposed model using English achieved an accuracy of 59.9%.

Baniata and Park (Baniata and Park 2016) proposed a combined CNN and Bi-LSTM in two different models using Arabic. The first model consisted of three convolutional layers with filter sizes of 3, 4, and 5, followed by the average pooling Bi-LSTM layer. The third layer was a merged layer that was connected to a fully connected layer including a multilayer perception and a sigmoid classifier. The second model was an inverse model starting with

the Bi-LSTM layer, which was connected with three convolutional layers with filter sizes of 3, 4 and 5 followed by a fully connected layer, which was a sigmoid classifier. The results demonstrated that the CNN-Bi-LSTM model achieved better sentence feature representation, with an accuracy of 86.43%, than the Bi-LSTM-CNN model, which achieved an accuracy of 66.26%. Table 9 shows the summary sentence/character-level sentiment analysis using hybrid neural networks for Arabic text.

Table 9. Sentence-level/character-level sentiment analysis using hybrid networks for Arabic text

Papers	Classifier	Sentiment Classification Level	Document/Text Representation	Dataset	Results
(Al-Azani and El-Alfy 2017).	LSTM, CNN-LSTM, Stacked LSTM, Combined-LSTM-SUM, Combined-LSTM-MUL, Combined-LSTM-CONC.	Sentence level.	Word2vec.	Two datasets of Arabic tweets: ASTD (Nabil et al. 2015) and ArTwitter(Abdulla et al. 2013).	LSTM-CONC with ArTwitter and skip-gram: 87.27%. LSTM-MUL with ASTD and CBOW: 81.63%.
(Alayba et al. 2018b).	CNN-LSTM.	Character level, character 5-gram level, and word level.	Row embedding based on the level.	Arabic Health Services Dataset (Alayba et al. 2017), ArTwitter(Abdulla et al. 2013), and ASTD (Nabil et al. 2015).	Main-AHS: 94.24%. Sub-AHS: 95.68% ArTwitter: 88.10% ASTD: 77.62%.
(González et al. 2017).	CNN-Bi-LSTM.	Sentence level.	Word2vec.	SemEval-2017 Task 4 (Rosenthal et al. 2017).	(task A): Arabic: 50.8%.
(Baniata and Park 2016).	CNN-Bi-LSTM, Bi-LSTM-CNN.	Sentence level.	Polyglot (Al-Rfou et al. 2013).	LABR (Aly and Atiya 2013).	CNN-Bi-LSTM: 86.43%.

As noted, using a CNN as the first layer could enhance the accuracy. Although there is a lack of character-level analysis, there are promising results. Moreover, Arabic pretrained word embedding strongly affects the analysis, which requires well-trained embedding at the word and character levels. Moreover, using word embedding concurrently with character embedding can enhance the accuracy, as shown in (El-Kilany et al. 2018).

Table 10 presents the reviewed algorithms, their advantages and disadvantages and assessments.

Table 10. Types of algorithms that have been reviewed in related works

Model type	Advantages	Disadvantages	Assessment
RNN	It can utilize the same transport function with the same parameters in each phase.	It has difficulty with training data, has high gradients at several phases, and it is not suitable for image or tabular datasets.	It is a type of artificial neural networks and is particularly helpful in translations and sentiment analysis.
SVM	It is the most efficient in high-dimensional areas or when the number of dimensions is greater than the number of samples, and it is relatively memory-efficient.	It is not suitable for large datasets and is not effective when the dataset contains much noise, so that the target groups overlap.	It is one of the most accurate algorithms in sentiment analysis, but it cannot be used to analyze large datasets.
LSTM	As one of the most successful RNN algorithms, it overcomes repeated network training problems and is capable of learning long-term dependencies.	It cannot be stacked in deep models, and it cannot track long-term dependencies.	The repeating module for LSTM is very complicated. Instead of a single neural network layer, there are four layers interacting in a special manner.
CRF	It is perfect for various segmentation and sequence tagging tasks such as sentiment analysis.	It is computationally complex in the training phase, and retraining is difficult when new training data samples are available.	It is a supervised machine learning algorithm; therefore, it requires a sufficiently large training sample.
GRU	GRU is superior to LSTM, as it can be trained in less time and more effectively. Moreover, it is simple to modify and does not require memory modules.	It cannot extract local context features.	It is an enhanced LSTM algorithm that has been improved in terms of the network architecture and effectiveness, but it does overcome the inherent defect of LSTM in capturing local text features.

6 Evaluation metrics

The following standard evaluation metrics have been used to measure the efficiency of the previously described approaches (Ruder et al. 2016a; Tamchyna and Veselovská 2016; Baniata and Park 2016; Abbes et al. 2017; Al-Azani and El-Alfy 2017; Baly et al. 2017a; González et al. 2017; Ponti et al. 2017; Abdullah and Shaikh 2018; Alayba et al. 2018b; Alhuzali et al. 2018; El-Kilany et al. 2018; Elnagar et al. 2018; Heikal et al. 2018; Samy et al. 2018; Wang and Lu 2018; Al-Azani and El-Alfy 2018; Abdou et al. 2018; Al-Smadi et al. 2018; Albadi et al. 2018; Abdullah et al. 2018; Alwehaibi and Roy 2018; Baccouche et al. 2018; Al-Smadi et al. 2019).

6.1 Accuracy

The accuracy is used to evaluate how often the model yields a correct classification, and it is defined as the percentage of correctly classified tweets out of the total number of tweets. Therefore, it can be represented mathematically as shown in Equation (16):

$$Accuracy = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{16}$$

6.2 Precision

The precision evaluates the strictness of the classifier output; it is defined as the percentage of tweets classified as positive correctly out of the total number of positive samples. It can be calculated as shown in Equation (17):

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{17}$$

6.3 Recall

We use the recall to measure the integrity of a classifier's output. It measures the percentage of actual tweets that were correctly classified and can be calculated as shown in Equation (18):

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{18}$$

6.4 F1 score

The F1 score combines the precision and recall as shown in Equation (19):

$$F1 = 2 * \frac{\mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{19}$$

In Equations (16)-(19), TP (true positive) is the number of tweets correctly classified as positive, TN (true negative) is the number of tweets correctly classified as negative, FP (false positive) is the number of tweets incorrectly classified as positive, and FN (false negative) is the number of tweets incorrectly classified as negative.

7 Research gaps

This section answers RQ4 by presenting the findings from the related studies. sentiment analysis is a challenging research area with diverse and complicated tasks. The main research directions focus on subjectivity categorization, sentiment categorization, lexicon creation, feature mining, feature sentiment classification, and attitude spam recognition.

Using RNNs for sentiment analysis has yielded accurate results, as these networks use previous sequential states to compute the current input, which is suitable for the natural language context (Elman 1990; Bengio et al. 1994; Kobayashi et al. 2010; Ayyadevara 2018; Souri et al. 2018).

We observed a lack of studies that used RNNs for Arabic sentiment analysis compared to other languages, such as English. As of 2018, the number of RNN English-language sentiment analysis studies reached 193, while it was only 24 studies for Arabic sentiment analysis. Moreover, a wider look at multilingual studies shows that in (González et al. 2017; Abdullah and Shaikh 2018; Abdullah et al. 2018), sentiments were analyzed in both Arabic

and English. However, in (Tamchyna and Veselovská 2016), sentiments were analyzed using seven languages: Arabic, Dutch, English, French, Russian, Spanish, and Turkish, whereas (Baccouche et al. 2018) analyzed data in three different languages: English, French, and Arabic. This finding demonstrates that using Arabic in this task is promising and still in its infancy, particularly for sentence-level tasks, where there are only six studies on Arabic aspect-based analysis and five studies on emotion analysis. Samy et al. (Samy et al. 2018) used GRU and C-GRU, but in (Alhuzali et al. 2018), a hybrid model was developed using supervised algorithms for automatically determining the emotion intensity and feelings for English and Arabic text. GRU was demonstrated to yield the best results, particularly in emotion detection. It was also found that the use of supervised algorithms with GRU yields an average F1 score of 70% on emotion detection compared with 60% by SVM-SGD. There was only one study on hate speech detection (Albadi et al. 2018), and there have not been studies on sarcasm or cyberbullying detection. This finding demonstrates the need for further research in that direction. Moreover, constructing deeper models may enhance the classification accuracy. For example, Al-Azani and El-Alfy (Al-Azani and El-Alfy 2017) used different architectures of LSTM networks, and the deeper architectures performed better than the simple LSTM. The denser layers being used in the case of deep learning models such as LSTM or GRUs try to obtain more enhanced features and a better relation that is not possible with the usual machine learning and state-of-the-art sentiment analysis algorithms. The time dependency factor of RNNs, as discussed in the previous sections, helps in determining the meaning of a user's text block or phrase, which makes these algorithms suitable for sentiment analysis.

The Arabic language can be challenging to work with, especially because of its semantics. RNNs, however, have achieved remarkable success in natural language processing and sentiment analysis. RNNs have connections between nodes to form a directed graph along a sequence. It is basically a sequence of neural network blocks linked to each other like a chain. Each message is passed to a successor. This property allows RNNs to exhibit temporal behavior and capture sequential data, which makes it a more natural approach when dealing with textual data such as that of the Arabic language since text is naturally sequential. RNNs cannot be used to process exceptionally long arrangements on the off chance that they utilize *tanh* as its initiation function. It is entirely flexible on the off chance that we use *ReLU* as the activation function.

In the case of GRU, tuning the hyperparameters is a difficult task that is also reflected while trying to change slightly in the already proven architectures. Even though many changes are made, the improvement in the accuracy is not significant enough. In the case of LSTM, the computation time is greater than that of the corresponding GRU structure. Hence, a tradeoff needs to be set up while choosing the best algorithm; this tradeoff depends on the problem statement and the kind of data.

A limited number of training and validation datasets have been either generated or organized from the Internet for both aspect-based and sentence-level sentiment analysis. More Arabic datasets must be created and analyzed. All of the existing research on aspect-level sentiment analysis achieved an accuracy of less than 90%; thus, tuning better models is a research opportunity. Hence, retraining the existing models that have been indicated to be superior and creating new architectures requires development and research. In sentence-level sentiment analysis, there is a lack of studies dealing with the analysis of emojis; only 3 such studies exist. Since emojis are represented syntactically as numbers and symbols, which are discarded in the preprocessing phase to simplify the analysis, emoji analysis holds a high potential of trend and sentiment indication, given the challenge of analyzing the exact sentiment.

Only one study (Albadi et al. 2018) explored hate speech detection in Arabic in addition to a unified dataset for this approach. Hate speech detection is one of the important directions of sentiment analysis, as social media makes it easy for anonymized opinions and

speeches to be shared, and there is an increasing likelihood of showing hate and aggression without accountability or liability (MacAvaney et al. 2019). The existing method has a low accuracy of only 79%, which suffers from a lack of interpretability and hence cannot be considered significant. Therefore, higher accuracies must be targeted to effectively improve the detection of hate speech. For sentiment classification levels, approximately 13 papers considered only sentence-level classification, and 6 papers considered aspect-based analysis. The models from (Elnagar et al. 2018; Alwehaibi and Roy 2018) have higher efficiency, while other models yield lower accuracy because of imbalanced datasets and due to focus on achieving a higher accuracy, regardless of the precision and recall, which shows that the proposed models are overfit. The discussed models must be tested on other datasets to check their performance consistency across all the datasets. On the other hand, the hybrid techniques use a mixture of both sentence-level and character-level analysis. In general, the accuracy of hybrid models is higher than that of the component models since they provide different perspectives for analysis, as shown in Table 9. Hence, further hybrid models are a research opportunity and are recommended for Arabic sentiment analysis studies. There are many research options that can be used while implementing these kinds of methodologies. Some of the evident scenarios for success address the tradeoff between the bias and the variance caused by a mixture of different types of methods. They not only try to fit the model to achieve a better prediction but also try to minimize the overfitting of the model in the training data, which makes the algorithm or the methodology poorly generalizable.

In addition, developing pretrained word and character embeddings enhances the accuracy of sentiment classification, covering the different types of feature engineering, such as part-of-speech, named entity recognition, and handling negation, for different dialects. Based on the study by (Alwehaibi and Roy 2018), character embedding has shown promising results, especially when combined with word embedding, as shown in (Al-Smadi et al. 2019), (Abdou et al. 2018), and (El-Kilany et al. 2018).

We observed that 55% of the datasets under study have fewer than 5,000 labeled data samples, 75% of which are unspecified dialects. This observation can be problematic in two ways: first, the small size of the datasets leads to less trustworthy results prone to overfitting, as shown in (Alayba et al. 2018b), with accuracies greater than 90%. Second, the undetermined dialects of most of them can result in a waste of effort that can accumulate from labeling each dataset with its dialect. These two observations are limitations to the research prosperity, and it is recommended that they be taken into consideration in future studies. Nevertheless, creating a unified open dataset for each dialect may help achieve fair comparisons. Moreover, creating a well-annotated dataset for different classification domains is required for the purpose of enhancing aspect-based analysis.

8 Conclusions and future work

In this paper, a systematic review of Arabic sentiment analysis using RNNs is presented. The four research questions covered neural network approaches for sentiment analysis, RNN approaches for sentiment analysis, related studies using RNNs for Arabic sentiment analysis, and research gaps.

It was also seen that the use of deep neural networks such as RNN-LSTM and GRUs can achieve enhanced performance in Arabic sentiment analysis compared with state-of-the-art sentiment analysis models using machine learning approaches. We also discussed in detail how the network architecture of these algorithms handles the information or the features and makes the prediction better. Although much research has been performed in the field of sentiment analysis for different languages, we found that Arabic sentiment analysis still needs many focus areas and concrete methodologies. To conduct proper research to achieve better results, a great deal of focus should also be given towards the fields of hate speech determination, emoji determination, etc. These methods can help not only retain the best

features describing the particular dataset but also preprocess the texts to create a clean and better corpus.

Future directions include the following: to develop Arabic word-embedding models that deal with negation words without ignoring the word structure. In addition, other types of models must be explored to identify the most suitable model for analyzing emojis and to enhance document- and aspect-level classification. Furthermore, several algorithms can be combined for efficient sentiment analysis in large datasets, since the accuracy of hybrid models is higher than that of the component models, as shown in Table 9. The hybrid model can be a combination of a CNN and an LSTM model and will retain the generation of an important feature that does not lose the semantics, meaning, or sequence of phrases and words that appeared in the near past and will appear again in the future. The hybrid model does require a large amount of computational power and much retraining and training time to reach the desired performance level.

The preprocessing area needs to be explored in Arabic sentiment analysis. It needs to be considered in detail when developing an approach that will boost the performance of the model being used. The hyperparameters, which play a large role in making the model better fit while avoiding overfitting, need to be a major part of the research area. The computational time of the algorithm, which depends to a great extent on the model execution time, should be considered in the analysis of the different hyperparameter tuning algorithms.

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